

Source Code of Underground Cable Fault Detection System

Python Implementation

```
# Python
def calculate_fault_distance(measured_voltage, supply_voltage=5.0, total_length_km=10.0):
    if measured_voltage < 0 or measured_voltage > supply_voltage:
        return {"status": "Error"}
    if measured_voltage == 0:
        return {"status": "Healthy"}
    voltage_ratio = measured_voltage / supply_voltage
    fault_distance = voltage_ratio * total_length_km
    return {"status": "FAULT DETECTED", "distance_km": round(fault_distance, 2)}
```

```
# Demo test readings: 0.0, 1.25, 2.5, 3.75, 5.0
```

C Implementation

```
// C
typedef struct { int status_code; float distance_km; } FaultResult;
FaultResult calculate_fault_distance(float measured_voltage, float supply_voltage, float total_length_km)
{
    FaultResult r;
    if(measured_voltage < 0 || measured_voltage > supply_voltage) {r.status_code = -1; return r;}
    if(measured_voltage == 0) {r.status_code = 0; r.distance_km = 0; return r;}
    r.distance_km = (measured_voltage / supply_voltage) * total_length_km;
    r.status_code = 1;
    return r;
}
```

Java Implementation

```
// Java
public class CableFaultLocator {
    public static FaultResult calculateFaultDistance(double measuredVoltage, double supplyVoltage, double totalLengthKm)
    {
        if(measuredVoltage < 0 || measuredVoltage > supplyVoltage) return new FaultResult("Error", null, "Invalid");
        if(measuredVoltage == 0.0) return new FaultResult("Healthy", null, "No fault");
        double d = (measuredVoltage / supplyVoltage) * totalLengthKm;
        return new FaultResult("FAULT DETECTED", d, "Fault located");
    }
}
```